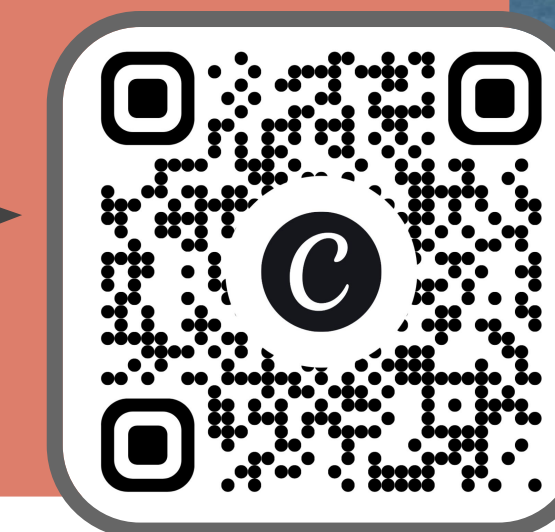




SEARCHING FOR AN ATMOSPHERE ON ROCKY EXOPLANET TOI-4481 B

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check out my website! →



Are we alone in the Universe?

A definitive answer to this question is likely far, far off in the future. *BUT!* JWST has allowed us to begin characterizing the atmospheres—or non-atmospheres—of rocky exoplanets in pursuit of that goal, particularly that of **M-Earths: terrestrial exoplanets orbiting M dwarf hosts**. It is thought that the *high levels of X-ray and ultraviolet irradiation* these planets receive from their host stars would *completely strip away their atmospheres*, and with the help of JWST we can now put that theory to the test. **TOI-4481 b** is today's test subject.

MEET TOI-4481 B

Rapid-fire stats of TOI-4481 b and its host star.^[1]

Planet	Star
• M_p : $1.90 \pm 0.17 M_\oplus$	★ M_* : $0.413 \pm 0.011 M_\odot$
• R_p : $1.331 \pm 0.023 R_\oplus$	★ R_* : $0.4144 \pm 0.0038 R_\odot$
• P: 0.9263237 ± 0.0000009 days	★ T_{eff} : 3600 ± 16 K
• a: 0.01406 ± 0.00030 AU	hypothetical visualization of TOI 4481 b ^[2] → 

- ## OUR DATA
- Three secondary eclipse observations with **MIRI LRS**, spanning roughly 5-12 μm
 - ◆ Shadowed region caused strong systematics beyond $\sim 10.5 \mu\text{m}$ in one observation; analysis limited to $< 10.5 \mu\text{m}$
 - Data reduction was done using the **Eureka!**^[3] pipeline
 - ◆ Eclipse depths were fit using **batman**^[4] and **emcee**^[5]
 - Stellar spectrum modeled with **pysynphot** using the **PHOENIX** library

WHY SO COLD QUESTION?

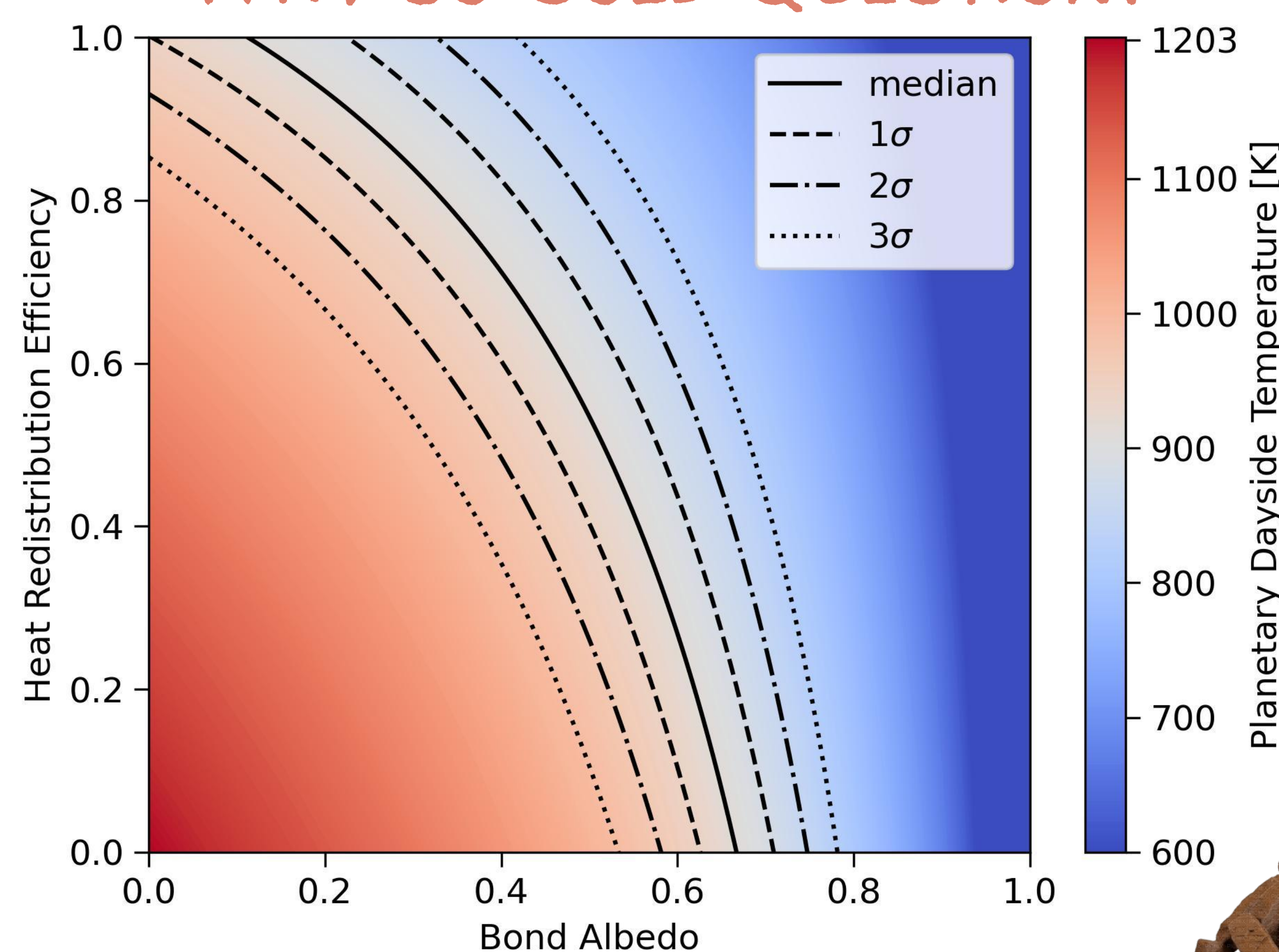


Figure 1. Contour plot showing our calculated planetary dayside temperature as a function of bond albedo and heat redistribution efficiency (a parameterization of the heat redistribution factor, higher efficiency → more heat transfer). This exemplifies that a shiny bare rock and a thick atmosphere could produce the same dayside temperature.

EMISSION SPECTRUM

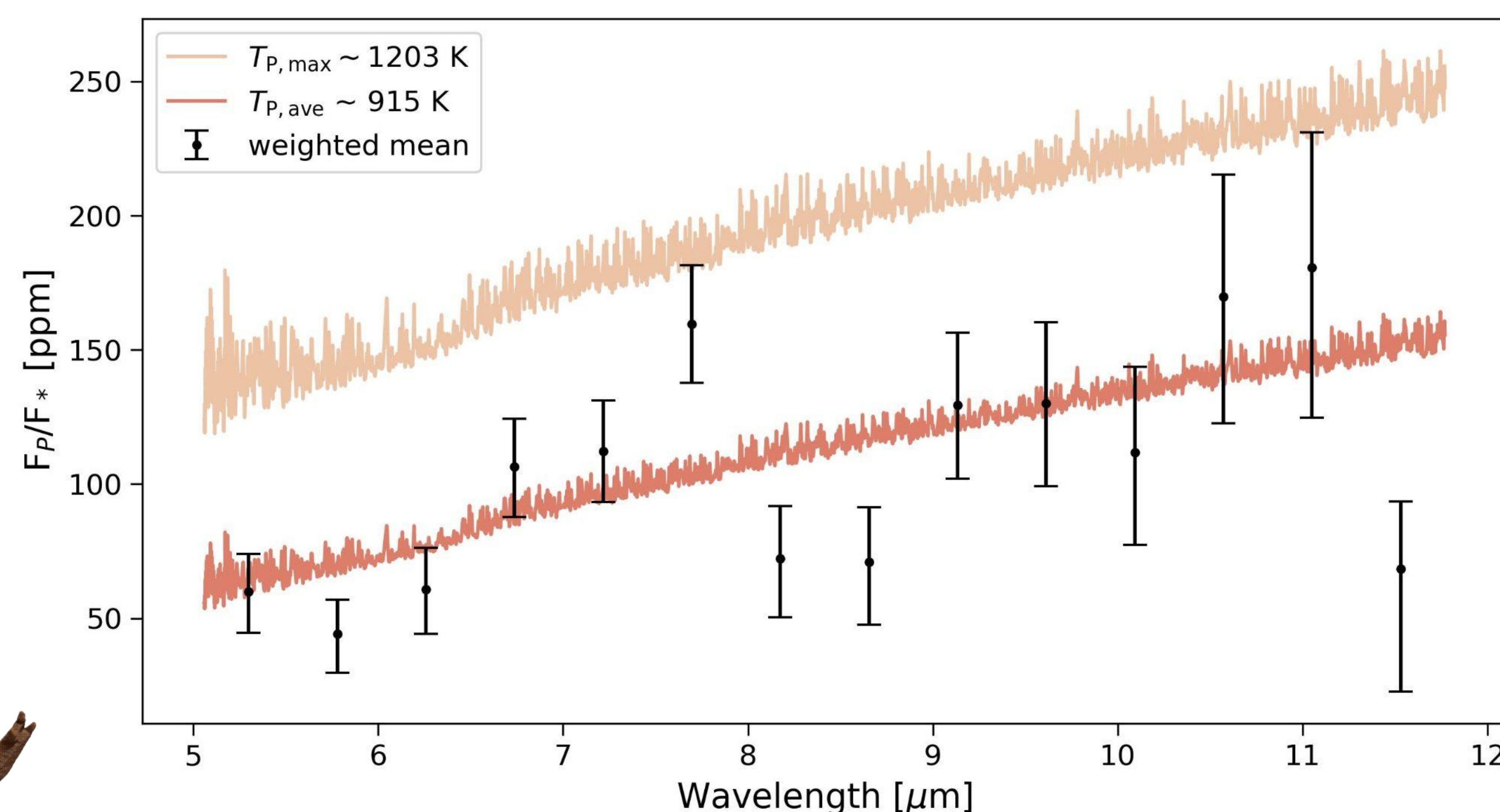


Figure 2. Error-weighted mean spectrum from our three datasets, with the exception of the points at ~ 11 and $\sim 11.5 \mu\text{m}$, which are only from the two datasets without the shadowed region. Blackbodies for the maximum dayside temperature and our calculated temperature are also plotted. Notably, our data does not strictly match either...atmosphere? TBD!

FINDINGS SO FAR...

- *Set-up:* when a planet goes behind its star, the total light we receive from the system decreases. We can measure this dip (known as the **eclipse depth**) → learn how much light the planet itself emits → calculate a dayside temperature for that planet.
- If TOI-4481 b was a **zero albedo, zero heat redistribution bare rock** → dayside temp. of **1203 ± 7 K**
- From our white-light eclipse depths, we calculate a dayside temperature of **$915 \pm 27/30$ K**
 - ◆ That may seem promising! However 🙅 there can be non-atmospheric tricks afoot.
 - A planet's dayside temperature depends on its *albedo* (A_B) and *heat redistribution factor* (f):

$$T_{\text{day}} = T_* (R_*/a)^{1/2} (1-A_B)^{1/4} f^{1/4}$$
- Example: a shiny (high albedo) bare rock can mimic a low-albedo planet with a thick atmosphere (lower heat redistribution factor)^[6] [Figure 1].
- Can break the degeneracy with **planetary emission spectra**: if blackbody → probably a bare rock, if features → maybe not? [Figure 2]
- ◆ Created by dividing the white-light curve into several narrower wavelength bins and fitting eclipse depths for each.

WHAT'S NEXT?

- Continue trying to recover the data in the shadowed region
- Additionally ensure that eclipse fits across all datasets are as robust as possible
- Create and compare models of various atmosphere and surface compositions to our emission spectrum
- Profit?

REFERENCES

[1] Palle et al. 2023, doi:10.1051/0004-6361/202244261

[2] <https://eyes.nasa.gov/apps/exo/#/>

[3] Bell et al. 2022, doi:10.21105/joss.04503

[4] Kreidberg 2025, doi:10.1086/683602

[5] Foreman-Mackey et al. 2013, doi:10.1086/670067

[6] Koll et al. 2019, doi:10.3847/1538-4357/ab4c91